CLASSIFICATION OF PLANT SEEDLINGS USING DEEP CNN ARCHITECTURES

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ABSTRACT-Weed management has a vital role in applications of agriculture domain. One of the key tasks is to identify the weeds after few days of plant germination which helps the farmers to perform early-stage weed management to reduce the contrary impacts on crop growth. Thus, we aim to classify the seedlings of crop and weed species. In this work, we propose a plant seedlings classification using the benchmark plant seedlings dataset. The dataset contains the images of 12 different species where three belongs to plant species and the other nine belongs to weed species. We implement the classification framework using three different deep convolutional neural network architectures, namely ResNet50V2, MobileNetV2 and EfficientNetB0. We train the models using transfer learning and compare the performance of each model on a test dataset of 833 images. We compare the three models and demonstrate that the EfficientNetB0 performs better with an average F1-Score of 96.26% and an accuracy of 96.52%.

Index terms—Convolutional Neural Network, Le Net, Alex Net.

I. INTRODUCTION

Precision farming is the revolution of traditional agriculture which focuses on crop production with controlled quality using evolving technologies. It focuses on the usage of drones, autonomous vehicles, robots and information technologies to achieve structured, sustainable, environment-friendly and cost-effective agriculture. Weed management is one of the key challenges of precision farming. Weeds are non-targeted plants that do not yield any profit to the farmers still compete for space and nutrients with the target crops which intern degrades the plant growth. Managing the weeds by human laborers is time-consuming and expensive. Even applying the herbicides uniformly across the farms will harm the unintended crops. Identifying and managing weeds based on their location and density helps to overcome the disadvantages of earlier approaches. Nowadays researchers have used the state of-

art algorithms of deep learning to implement many agricultural applications. Even we have used the deep CNNs to implement the plant seedlings classification. Many researchers have proposed frameworks for weeds and plant seedlings classification using a transfer learning approach and CNNs. Authors in [1] have implemented maize and weed classification using Le Net, Alex Net, c NET and s NET architectures and used outperformed c NET for real-time implementation. Authors in [2] have implemented a framework to classify weeds of Australian rangelands using pre-trained ResNet50 and InceptionV3 architectures and proposed a real-time robotic weed control system using outperformed pre-trained ResNet50. Authors in [3] have proposed the Philippine Indigenous plant seedlings classification frameworks by fine-tuning pre-trained Alex Net Google Net and ResNet50 architectures. ResNet50 performs better over the other two architectures. Authors in [4] have implemented a carrot and weed classifier using the CNN.

II. LITERATURE SURVEY

Precise weed and maize classification through convolutional neuronal networks

Deep Learning is playing an important role in big data processing for more accurate modeling of common productive processes. It is being widely used in artificial vision applications and specifically in pattern recognition. The versatility of deep learning has positioned it as a fit tool used in many fields of application, among which is precision agriculture. This paper presents the development of an algorithm capable of image segmentation and classification. Segmentation is intended to separate the target plant from the original image, while classification is meant to identify what images belong to the two defined classes. It applies a convolutional neural network (CNN) to discriminate maize plants from weeds in real time, at early crop development stages. It was applied to maize crop because it is a common staple crop in the Ecuadorian Highlands. The convolutional neural network has been trained with a dataset generated in the segmentation stage. The performance of the network was analyzed with LeNET, AlexNet, cNET and sNET network architectures. The network architecture that presented the best training results was cNET based on its performance in terms of accuracy and processing time. The minimum working filter number for this network architecture was 16. The best performing algorithms and processors have a significant potential for autonomous weed and

crop classification systems in a real-time application.

Philippine Indigenous Plant Seedlings Classification Using Deep Learning

Plant taxonomists have specialized knowledge on a specific plant species. The shortage of these experts and their unbalanced distribution throughout the globe remains to be a serious problem worldwide. Deep learning technology can be used to create tools that will help them speed up the process of plant classification. This study implements three deep learning models to classify images of Philippine indigenous plant seedlings into five species. AlexNet, GoogLeNet, and ResNet50 were fine-tuned for this purpose. In the three pre-trained models, the weight and bias learning rate factors of the fully connected layers were both increased to 20 to speed up learning.

III. PROPOSED SYSTEM

The overview of our proposed system is shown in the below figure.

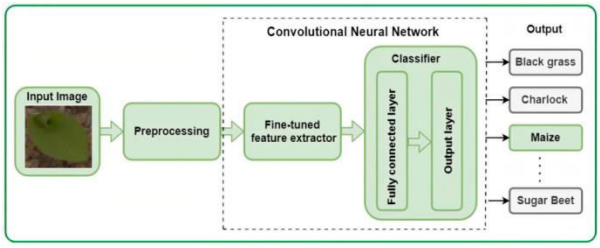

Fig. 1: System Overview

Implementation Modules

Dataset

- In this module, we collect the plant seedlings image dataset from internet. We take the all the plant seedlings. The images were available in three formats .jpg, .pngand .jpeg. This Image data is not enough for train the model and test the model. To increase the data size we perform data agumentation step in next level.

Data Augmentation

- Data Augmentation is the process of increasing the size of the data set. There are ways in which the process is done by rotating, flipping, shearing, and adding random noise, along with other types. The new images in the dataset will help in training the network as well as increasing the efficiency of classifying the testing data or the new data.

Preprocessing

- In this module, we pre-process the image data and convert the image data into numpy array data. This step is very important to identify the feature of the image data. This extracted features are show as array data and size.

Train Model

- In this module, after spilt data as train and test data in the ratio of 80% and 20%

respectively. The train data can be used for train the model and the test data can be used for test the model performance. In this project we applied CNN Model and to train the model we are using fit() method in python programming.

Classification

In this module, we used our proposed model to classify the plant seedlings.

Implementation Algorithms

Convolutional Neural Network (CNN)

In Convolutional Neural Network model, the input image is convolved through a filter collection in the convolution layers to produce a feature map. Each feature map is then combined to fully connected networks, and the face expression is recognized to belong to a particular class-based output of the softmax algorithm

IV. RESULTS

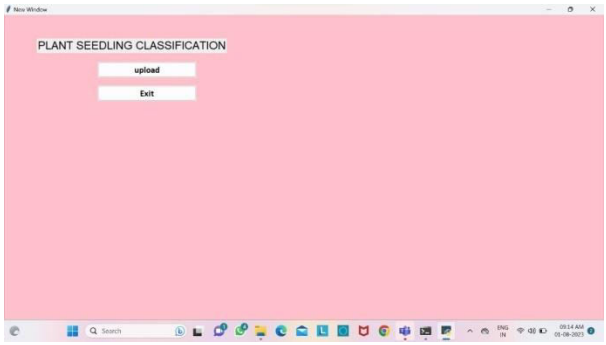

Fig. 2: Home Page

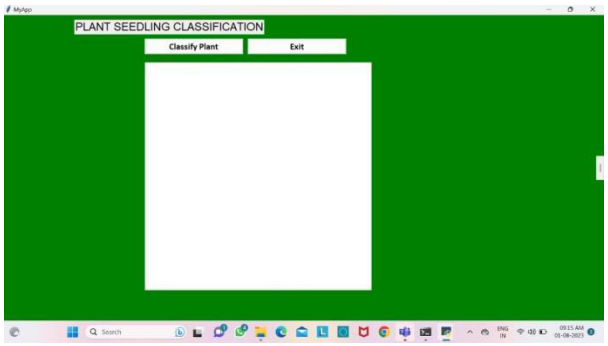

Fig. 3: Classification Page

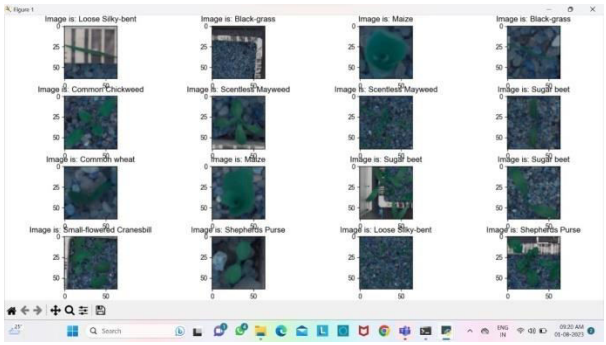

Fig. 4: Image Dataset

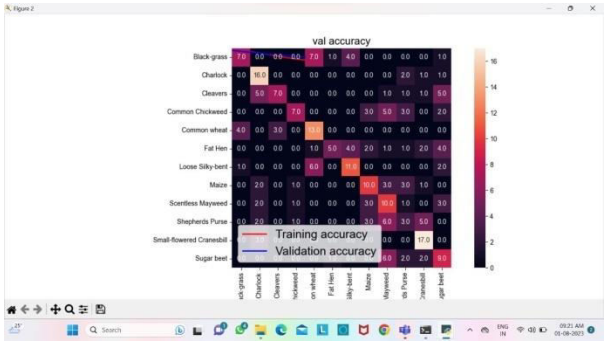

Fig. 4: Heat Graph

Fig. 5: Comparison Graph

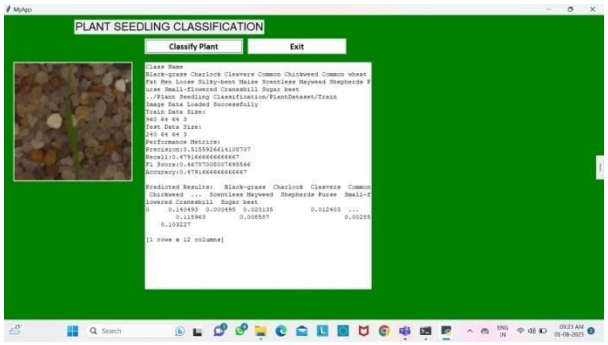

Fig. 5: Result Page

V. CONCLUSION

We proposed a plant seedlings classification framework using ResNet50V2, MobileNetV2 and EfficientNetB0 architectures. The models are validated using the benchmark plant seedlings dataset, which contains 12 different species where three belongs to plant species and the other nine belongs to weed species. We compared the models and demonstrated that the EfficientNetB0 model outperformed with an average F1-Score of 96.26% and an accuracy of 96.52%.

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